Unit 3, Lesson 7: Solving Real-World Problems with Rational Numbers – Part 3

Benchmark

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

Learning Target

- ☐ I can solve problems with three operations involving any of the four operations with rational numbers.

3.7.1 Warm-Up: Bell Work

1. Number each operation using numbers 1-4 to reflect the correct order of the order of operations.

◯	Perform addition and subtraction in order from left to right. (→)
◯	Perform operations inside grouping symbols.
◯	Evaluate expressions with exponents.
◯	Perform multiplication and division in order from left to right. (→)

3.7.2 Activity: Error Analysis

With your partner, answer each question below.

1. Diana was given the following problem to solve.

   Eugene has $\$40.00$ to spend at a theme park gift shop. He receives a discount of $\frac{1}{5}$ off the price of an item for being an annual passholder. He wants to purchase a collectible passholder pin that has a retail price of $\$19.95$. How much will he have left after purchasing the pin?

   Diana solved the problem with the steps shown.

Diana's Work
$19.95 \cdot \left(-\frac{1}{5}\right)$ $\frac{19.95}{1} \cdot \left(-\frac{1}{5}\right)$ $-\frac{19.95}{5}$ $5\overline{)19.95} = 3.99$: -15, 49, -45, 45, -45, 0 $\$40.00 - \$3.99$ $40.00 - 3.99 = 36.01$ Eugene had \$36.01 left.

   a. Identify what Diana did correctly and incorrectly.

Correct	Incorrect

b. How much money will Eugene have left over after buying the pin? Show your work.

2. Jeffrey was given the following problem to solve.

Diego is planning to hike the very difficult Shorty's Well Route to Telescope Peak, which begins at an elevation of -262 feet and ends at an elevation of $11{,}043$ feet. He wants to hike $\frac{3}{5}$ of the change in elevation of the trail on the first day of his hike. If Diego sticks to his plan, at what elevation will he start on the second day?

Jeffrey solved the problem with the steps shown.

Jeffrey's Work

$11043 - 262 =$

$$\begin{array}{r} 1\cancel{1}\cancel{0}43 \\ -\ \ 262 \\ \hline 10781 \end{array} \text{ feet}$$

$$\frac{10781}{1} \cdot \frac{3}{5} = \frac{32343}{5}$$

$$\begin{array}{r} 6468.6 \text{ feet} \\ 5\overline{)32343.0} \\ -30 \\ \hline 23 \\ -20 \\ \hline 34 \\ -30 \\ \hline 43 \\ -40 \\ \hline 30 \\ -30 \\ \hline 0 \end{array}$$

Diego plans to be at 6468.6 feet to start Day 2.

a. Identify what Jeffrey did correctly and incorrectly.

Correct	Incorrect

b. At what elevation will Diego start on the second day? Show your work.

3. Ashlyn was given the following problem to solve.

Fernando runs after school to prepare for the cross-country season. On Monday he ran 4.34 miles, Wednesday he ran 6.56 miles, and on Friday he ran 1.375 miles. His teammate Marcus only ran $\frac{4}{10}$ the number of miles as Fernando. How many miles did Marcus run?

Ashlyn solved the problem with the steps shown.

Ashlyn's Work

$$\begin{array}{r} \overset{1}{4}.34 \\ +6.56 \\ \hline 10.90 \end{array}$$

$$23.65 \div \frac{4}{10}$$

$$\frac{23.65}{1} \cdot \frac{10}{4} = \frac{236.5}{4}$$

$$\begin{array}{r} 59.0 \\ 4\overline{)236.5} \\ -20 \\ \hline 36 \\ -36 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 10.90 \\ +1.375 \\ \hline 23.65 \end{array}$$ miles

Fernando ran 59 miles.

a. Identify what Ashlyn did correctly and incorrectly.

Correct	Incorrect

b. How many miles did Marcus run? Show your work.

4. Craig was given the following problem to solve.

Rhonda makes a specific hair dye color by mixing $1\frac{1}{8}$ ounces (oz) of red tone, 0.75 oz of gray tone, and $\frac{5}{8}$ oz of brown tone. If Rhonda needs to make 20 oz of the same color dye, how much of each color tone is needed?

Craig solved the problem with the steps shown.

Craig's Work

a. Identify what Craig did correctly and incorrectly.

Correct	**Incorrect**

b. How much of each color is needed to make 20 oz of dye? Show your work.

3.7.3 Collaboration: Finding Missing Values

Many homeowners have solar panels that can generate energy to power their homes. The Better Utility Company charges customers who have solar panels using the following criteria:

- Customers are charged $\$0.12$ per kilowatt-hour for the energy used that is provided by the utility company.
- The utility company does not charge customers for energy they use from their own solar panel(s).
- For every kilowatt-hour of electricity that customers with solar panels generate but do not use, the utility company gives a price adjustment of $-\$0.025$.

Erin Hill's bill for the month of July is missing some information.

Best Utility Company 123 Ohms Way Big City, ST 12345		Erin Hill 456 Long St. Small Town, ST 98765	
Kilowatt-hours used from Best Utility Company	________	Charge	\$82.04
Kilowatt-hours generated by solar panel and not used by customer	________	Credit	\$−4.10
		Amount Due	\$________

1. Work with your partner to fill in the missing information on the bill. Round answers to nearest hundredth.

2. Find another classmate other than your partner and compare your missing values.

   - If the missing values for both are the same, have your classmate initial next to each value in your bill.
   - If the missing values are different, discuss your work to determine what error(s) were made. After correcting any mistakes, initial next to each value in each other's bill.

3.7.4 Wrap-Up: Reflection

1. Complete the table by reflecting on the day's lesson.

Today I deserve  for …	
Today I  by doing …	
Today I was  by …	

